



Банк России

УЧЕТ ЛАГОВ В МОДЕЛИРОВАНИИ ВРЕМЕННЫХ РЯДОВ

АРЖЕНОВСКИЙ СЕРГЕЙ,
ЭО РОСТОВ-НА-ДОНУ

НОЯБРЬ 2025





Динамические модели (с авторегрессионно распределенным запаздыванием, Autoregressive Distributed Lag models – ADL)

Модель в общем виде выглядит так:

$$y_t = a_0 + a_1 y_{t-1} + \dots + a_p y_{t-p} + (\beta_{10} x_{1t} + \beta_{11} x_{1t-1} + \dots + \beta_{1r} x_{1t-r}) + \dots \\ \dots + (\beta_{s0} x_{st} + \beta_{s1} x_{st-1} + \dots + \beta_{sr} x_{st-r}) + \varepsilon_t,$$

p – глубина запаздывания по переменной y_t ,

r – глубина запаздывания по переменным $x_{1t}, \dots, x_{st}$,

s – количество экзогенных переменных.

Обозначение: ADL(p, r, s).



Модель ADL(1, 1, 1)

Модель выглядит так:

$$y_t = \mu + a_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t. \quad (1)$$

Различные виды моделей из (1):

1. Статическая регрессия ($a_1 = \beta_1 = 0$):

$$y_t = \mu + \beta_0 x_t + \varepsilon_t.$$

2. Процесс авторегрессии ($\beta_0 = \beta_1 = 0$):

$$y_t = \mu + a_1 y_{t-1} + \varepsilon_t.$$

3. Модель опережающего показателя ($a_1 = \beta_0 = 0$):

$$y_t = \mu + \beta_1 x_{t-1} + \varepsilon_t.$$

4. Модель скорости роста ($a_1 = 1, \beta_1 = -\beta_0$):

$$\Delta y_t = \mu + \beta_0 \Delta x_t + \varepsilon_t.$$



Модель ADL(1, 1, 1)

5. Модель распределенных запаздываний ($a_1=0$):

$$y_t = \mu + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t.$$

6. Модель частичной корректировки ($\beta_1=0$):

$$y_t = \mu + a_1 y_{t-1} + \beta_0 x_t + \varepsilon_t.$$

7. Фальстарт (или приведенная форма) ($\beta_0=0$):

$$y_t = \mu + a_1 y_{t-1} + \beta_1 x_{t-1} + \varepsilon_t.$$

8. Авторегрессионные ошибки ($\beta_1 = -a_1\beta_0$):

$$y_t = \mu + a_1 y_{t-1} + \beta_0 x_t - a_1 \beta_0 x_{t-1} + \varepsilon_t \text{ или } y_t - a_1 y_{t-1} = (1 - a_1)\mu + \beta_0(x_t - a_1 x_{t-1}) + \varepsilon_t.$$

9. Модель коррекции ошибок ($|a_1| < 1, \beta_0 + \beta_1 = b(1 - a_1), b \neq 0$):

$$\Delta y_t = \mu + \beta_0 \Delta x_t - (1 - a_1)(y_{t-1} - b x_{t-1}) + \varepsilon_t.$$



Моделирование в Eviews

Пусть известна модель, которой следуют данные (DGP):

$$\begin{aligned}y_t &= 0,5y_{t-1} + 0,2x_t + 0,3x_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0; 0,1^2), \\x_t &= 0,5x_{t-1} + v_t, \quad v_t \sim N(0; 0,5^2).\end{aligned}$$

Получим значения переменных y_t и x_t , которые следуют указанной модели в Eviews:

```
smpl @first @first  
series x=0  
series y=0  
smpl @first+1 @last  
series x=0.5*x(-1)+0.5*nrnd  
series y=0.5*y(-1)+0.2*x+0.3*x(-1)+0.1*nrnd  
smpl @all
```



Моделирование в Eviews

0. Построим МНК модель вида:

$$y_t = \mu + a_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$$

С константой

Dependent Variable: Y
Method: Least Squares
Date: 09/10/25 Time: 17:19
Sample (adjusted): 2 100
Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.001415	0.010666	0.132673	0.8947
Y(-1)	0.454715	0.034687	13.10914	0.0000
X	0.196657	0.021719	9.054607	0.0000
X(-1)	0.312661	0.027672	11.29884	0.0000
R-squared	0.939631	Mean dependent var	0.089555	
Adjusted R-squared	0.937725	S.D. dependent var	0.414010	
S.E. of regression	0.103316	Akaike info criterion	-1.662477	
Sum squared resid	1.014055	Schwarz criterion	-1.557624	
Log likelihood	86.29264	Hannan-Quinn criter.	-1.620054	
F-statistic	492.8842	Durbin-Watson stat	2.261603	
Prob(F-statistic)	0.000000			

Без константы

Dependent Variable: Y
Method: Least Squares
Date: 09/10/25 Time: 17:21
Sample (adjusted): 2 100
Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Y(-1)	0.455237	0.034286	13.27771	0.0000
X	0.196979	0.021472	9.173687	0.0000
X(-1)	0.312633	0.027529	11.35642	0.0000
R-squared	0.939620	Mean dependent var	0.089555	
Adjusted R-squared	0.938362	S.D. dependent var	0.414010	
S.E. of regression	0.102786	Akaike info criterion	-1.682494	
Sum squared resid	1.014243	Schwarz criterion	-1.603854	
Log likelihood	86.28346	Hannan-Quinn criter.	-1.650676	
Durbin-Watson stat	2.262536			

Коррелограмма: «белый шум».

Тест Бройша-Годфри: а/к нет ($p=0,14$, AR(1)).

Тест на гомоскедастичность Уайта: $p=0,42$.

Нормальность распр. остатков: $p=0,37$.



Моделирование в Eviews

1. Статическая регрессия:

$$y_t = \mu + \beta_0 x_t + \varepsilon_t.$$

Dependent Variable: Y
Method: Least Squares
Date: 09/10/25 Time: 17:46
Sample: 1 100
Included observations: 100

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.036612	0.030935	1.183514	0.2395
X	0.482324	0.052705	9.151465	0.0000
R-squared	0.460796	Mean dependent var		0.088660
Adjusted R-squared	0.455294	S.D. dependent var		0.412011
S.E. of regression	0.304081	Akaike info criterion		0.476754
Sum squared resid	9.061609	Schwarz criterion		0.528857
Log likelihood	-21.83768	Hannan-Quinn criter.		0.497841
F-statistic	83.74932	Durbin-Watson stat		1.091221
Prob(F-statistic)	0.000000			



Моделирование в Eviews

2. Авторегрессия:

$$y_t = \mu + a_1 y_{t-1} + \varepsilon_t.$$

Dependent Variable: Y
Method: Least Squares
Date: 09/11/25 Time: 16:33
Sample (adjusted): 2 100
Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.021844	0.023757	0.919468	0.3601
Y(-1)	0.851277	0.058003	14.67643	0.0000
R-squared	0.689498	Mean dependent var	0.089555	
Adjusted R-squared	0.686297	S.D. dependent var	0.414010	
S.E. of regression	0.231883	Akaike info criterion	-0.065169	
Sum squared resid	5.215679	Schwarz criterion	-0.012743	
Log likelihood	5.225888	Hannan-Quinn criter.	-0.043958	
F-statistic	215.3976	Durbin-Watson stat	1.391399	
Prob(F-statistic)	0.000000			

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 1 lag

F-statistic	13.74276	Prob. F(1,96)	0.0004
Obs*R-squared	12.39748	Prob. Chi-Square(1)	0.0004



Моделирование в Eviews

3. Модель опережающего показателя:

$$y_t = \mu + \beta_1 x_{t-1} + \varepsilon_t.$$

Dependent Variable: Y
 Method: Least Squares
 Date: 09/11/25 Time: 16:58
 Sample (adjusted): 2 100
 Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.027161	0.019741	1.375879	0.1720
X(-1)	0.638848	0.034093	18.73837	0.0000

R-squared	0.783543	Mean dependent var	0.089555
Adjusted R-squared	0.781312	S.D. dependent var	0.414010
S.E. of regression	0.193608	Akaike info criterion	-0.425969
Sum squared resid	3.635949	Schwarz criterion	-0.373542
Log likelihood	23.08545	Hannan-Quinn criter.	-0.404757
F-statistic	351.1266	Durbin-Watson stat	1.607038
Prob(F-statistic)	0.000000		

Breusch-Godfrey Serial Correlation LM Test:
 Null hypothesis: No serial correlation at up to 1 lag

F-statistic	4.009759	Prob. F(1,96)	0.0481
Obs*R-squared	3.969274	Prob. Chi-Square(1)	0.0463



Моделирование в Eviews

4. Модель скорости роста:

$$\Delta y_t = \mu + \beta_0 \Delta x_t + \varepsilon_t.$$

Dependent Variable: D(Y)
 Method: Least Squares
 Date: 09/11/25 Time: 17:07
 Sample (adjusted): 2 100
 Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.009060	0.023646	0.383157	0.7024
D(X)	0.084224	0.044049	1.912062	0.0588
R-squared	0.036322	Mean dependent var		0.010015
Adjusted R-squared	0.026387	S.D. dependent var		0.238387
S.E. of regression	0.235221	Akaike info criterion		-0.036588
Sum squared resid	5.366903	Schwarz criterion		0.015839
Log likelihood	3.811094	Hannan-Quinn criter.		-0.015376
F-statistic	3.655981	Durbin-Watson stat		1.678507
Prob(F-statistic)	0.058819			

Date: 09/11/25 Time: 17:09
 Sample: 1 100
 Included observations: 99
 Q-statistic probabilities adjusted for 1 dynamic regressor

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*	
		1	0.156	0.156	2.4674	0.116
		2	-0.019	-0.044	2.5047	0.286
		3	-0.290	-0.288	11.273	0.010
		4	-0.183	-0.106	14.807	0.005
		5	-0.084	-0.057	15.557	0.008
		6	-0.068	-0.151	16.057	0.013
		7	-0.060	-0.139	16.454	0.021
		8	0.064	0.023	16.910	0.031
		9	0.041	-0.062	17.099	0.047
		10	0.122	0.032	18.771	0.043

*Probabilities may not be valid for this equation specification.

Breusch-Godfrey Serial Correlation LM Test:
 Null hypothesis: No serial correlation at up to 3 lags

F-statistic	3.850566	Prob. F(3,94)	0.0120
Obs*R-squared	10.83467	Prob. Chi-Square(3)	0.0127



Моделирование в Eviews

5. Модель распределенных запаздываний:

$$y_t = \mu + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t.$$

Dependent Variable: Y

Method: Least Squares

Date: 09/12/25 Time: 11:47

Sample (adjusted): 2 100

Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.017288	0.017668	0.978511	0.3303
X	0.186433	0.036187	5.151878	0.0000
X(-1)	0.531873	0.036759	14.46915	0.0000

R-squared	0.830427	Mean dependent var	0.089555
Adjusted R-squared	0.826894	S.D. dependent var	0.414010
S.E. of regression	0.172253	Akaike info criterion	-0.649871
Sum squared resid	2.848424	Schwarz criterion	-0.571231
Log likelihood	35.16861	Hannan-Quinn criter.	-0.618053
F-statistic	235.0632	Durbin-Watson stat	1.115164
Prob(F-statistic)	0.000000		

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 1 lag

F-statistic	22.20543	Prob. F(1,95)	0.0000
Obs*R-squared	18.75628	Prob. Chi-Square(1)	0.0000



Моделирование в Eviews

6. Модель частичной корректировки:

$$y_t = \mu + a_1 y_{t-1} + \beta_0 x_t + \varepsilon_t.$$

Dependent Variable: Y
 Method: Least Squares
 Date: 09/12/25 Time: 11:53
 Sample (adjusted): 2 100
 Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000502	0.016244	0.030901	0.9754
Y(-1)	0.691551	0.042090	16.43021	0.0000
X	0.312358	0.029170	10.70822	0.0000
R-squared	0.858505	Mean dependent var		0.089555
Adjusted R-squared	0.855557	S.D. dependent var		0.414010
S.E. of regression	0.157347	Akaike info criterion		-0.830893
Sum squared resid	2.376773	Schwarz criterion		-0.752253
Log likelihood	44.12921	Hannan-Quinn criter.		-0.799075
F-statistic	291.2348	Durbin-Watson stat		2.618479
Prob(F-statistic)	0.000000			

Breusch-Godfrey Serial Correlation LM Test:
 Null hypothesis: No serial correlation at up to 1 lag

F-statistic	12.84324	Prob. F(1,95)	0.0005
Obs*R-squared	11.79008	Prob. Chi-Square(1)	0.0006



Моделирование в Eviews

7. Приведенная форма:

Dependent Variable: Y
Method: Least Squares
Date: 09/12/25 Time: 12:17
Sample (adjusted): 2 100
Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.012210	0.014392	0.848410	0.3983
Y(-1)	0.443437	0.047067	9.421353	0.0000
X(-1)	0.430795	0.033134	13.00140	0.0000
R-squared	0.887532	Mean dependent var	0.089555	
Adjusted R-squared	0.885189	S.D. dependent var	0.414010	
S.E. of regression	0.140282	Akaike info criterion	-1.060486	
Sum squared resid	1.889194	Schwarz criterion	-0.981846	
Log likelihood	55.49407	Hannan-Quinn criter.	-1.028668	
F-statistic	378.7872	Durbin-Watson stat	2.118153	
Prob(F-statistic)	0.000000			

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

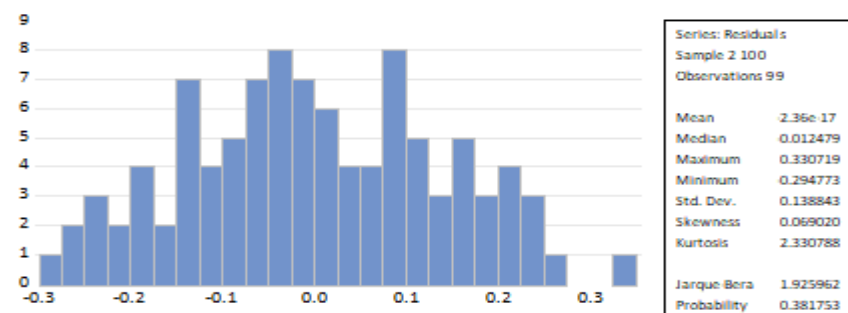
F-statistic	0.518356	Prob. F(5,93)	0.7618
Obs*R-squared	2.684189	Prob. Chi-Square(5)	0.7485
Scaled explained SS	1.679438	Prob. Chi-Square(5)	0.8915

$$y_t = \mu + a_1 y_{t-1} + \beta_1 x_{t-1} + \varepsilon_t.$$

Date: 09/12/25 Time: 12:19
Sample: 1 100
Included observations: 99
Q-statistic probabilities adjusted for 1 dynamic regressor

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
1		-0.071	-0.071	0.5178	0.472
2		0.044	0.040	0.7216	0.697
3		0.088	0.094	1.5241	0.677
4		-0.102	-0.093	2.6280	0.622
5		-0.096	-0.120	3.6090	0.607
6		-0.048	-0.063	3.8543	0.696
7		-0.067	-0.048	4.3455	0.739
8		0.085	0.096	5.1429	0.742
9		-0.037	-0.029	5.2957	0.808
10		-0.047	-0.078	5.5388	0.852

*Probabilities may not be valid for this equation specification.





Моделирование в Eviews

7. Приведенная форма:

$$y_t = \mu + a_1 y_{t-1} + \beta_1 x_{t-1} + \varepsilon_t.$$

Dependent Variable: Y
Method: Least Squares
Date: 09/12/25 Time: 12:25
Sample (adjusted): 2 100
Included observations: 99 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Y(-1)	0.447840	0.046713	9.587147	0.0000
X(-1)	0.432242	0.033043	13.08132	0.0000
R-squared	0.886688	Mean dependent var	0.089555	
Adjusted R-squared	0.885520	S.D. dependent var	0.414010	
S.E. of regression	0.140079	Akaike info criterion	-1.073218	
Sum squared resid	1.903359	Schwarz criterion	-1.020792	
Log likelihood	55.12431	Hannan-Quinn criter.	-1.052006	
Durbin-Watson stat	2.119237			



Моделирование в Eviews

8. Авторегрессионные ошибки:

$$y_t = \mu + a_1 y_{t-1} + \beta_0 x_t - a_1 \beta_0 x_{t-1} + \varepsilon_t.$$

Dependent Variable: Y

Method: Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/12/25 Time: 13:51

Sample (adjusted): 2 100

Included observations: 99 after adjustments

Convergence achieved after 34 iterations

Coefficient covariance computed using outer product of gradients

Y = C(1)+C(2)*Y(-1)+C(3)*X-C(2)*C(3)*X(-1)

	Coefficient	Std. Error	t-Statistic	Prob.
C(1)	0.021316	0.023335	0.913478	0.3633
C(2)	0.818927	0.063844	12.82695	0.0000
C(3)	0.106873	0.046623	2.292282	0.0241

R-squared	0.705227	Mean dependent var	0.089555
Adjusted R-squared	0.699086	S.D. dependent var	0.414010
S.E. of regression	0.227108	Akaike info criterion	-0.096951
Sum squared resid	4.951476	Schwarz criterion	-0.018311
Log likelihood	7.799080	Hannan-Quinn criter.	-0.065133
F-statistic	114.8371	Durbin-Watson stat	1.580603
Prob(F-statistic)	0.000000		

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 1 lag

F-statistic	6.272393	Prob. F(1,95)	0.0140
Obs*R-squared	6.131650	Prob. Chi-Square(1)	0.0133



Моделирование в Eviews

9. Модель коррекции ошибок:

$$\Delta y_t = \mu + \beta_0 \Delta x_t - (1 - a_1)(y_{t-1} - bx_{t-1}) + \varepsilon_t.$$

Dependent Variable: D(Y)

Method: Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/12/25 Time: 15:06

Sample (adjusted): 2 100

Included observations: 99 after adjustments

Convergence achieved after 4 iterations

Coefficient covariance computed using outer product of gradients

D(Y) = C(1)+C(2)*D(X)+(C(3)-1)*(Y(-1)-C(4)*X(-1))

	Coefficient	Std. Error	t-Statistic	Prob.
C(1)	0.001415	0.010666	0.132673	0.8947
C(2)	0.196657	0.021719	9.054607	0.0000
C(3)	0.454715	0.034687	13.10914	0.0000
C(4)	0.934039	0.048082	19.42612	0.0000

R-squared	0.817917	Mean dependent var	0.010015
Adjusted R-squared	0.812167	S.D. dependent var	0.238387
S.E. of regression	0.103316	Akaike info criterion	-1.662477
Sum squared resid	1.014055	Schwarz criterion	-1.557624
Log likelihood	86.29264	Hannan-Quinn criter.	-1.620054
F-statistic	142.2465	Durbin-Watson stat	2.261603
Prob(F-statistic)	0.000000		

Dependent Variable: D(Y)

Method: Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/12/25 Time: 15:09

Sample (adjusted): 2 100

Included observations: 99 after adjustments

Convergence achieved after 3 iterations

Coefficient covariance computed using outer product of gradients

D(Y) = C(2)*D(X)+(C(3)-1)*(Y(-1)-C(4)*X(-1))

	Coefficient	Std. Error	t-Statistic	Prob.
C(2)	0.196979	0.021472	9.173687	0.0000
C(3)	0.455237	0.034286	13.27771	0.0000
C(4)	0.935475	0.046711	20.02700	0.0000

R-squared	0.817883	Mean dependent var	0.010015
Adjusted R-squared	0.814089	S.D. dependent var	0.238387
S.E. of regression	0.102786	Akaike info criterion	-1.682494
Sum squared resid	1.014243	Schwarz criterion	-1.603854
Log likelihood	86.28346	Hannan-Quinn criter.	-1.650676
Durbin-Watson stat	2.262536		

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 1 lag

F-statistic	2.254379	Prob. F(1,95)	0.1366
Obs*R-squared	2.294843	Prob. Chi-Square(1)	0.1298



Моделирование в Eviews

9. Модель коррекции ошибок:

$$\Delta y_t = \beta_0 \Delta x_t - (1 - a_1)(y_{t-1} - b x_{t-1}) + \varepsilon_t.$$

Dependent Variable: D(Y)

Method: Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/12/25 Time: 15:09

Sample (adjusted): 2 100

Included observations: 99 after adjustments

Convergence achieved after 3 iterations

Coefficient covariance computed using outer product of gradients

D(Y) = C(2)*D(X)+(C(3)-1)*(Y(-1)-C(4)*X(-1))

	Coefficient	Std. Error	t-Statistic	Prob.
C(2)	0.196979	0.021472	9.173687	0.0000
C(3)	0.455237	0.034286	13.27771	0.0000
C(4)	0.935475	0.046711	20.02700	0.0000
R-squared	0.817883	Mean dependent var	0.010015	
Adjusted R-squared	0.814089	S.D. dependent var	0.238387	
S.E. of regression	0.102786	Akaike info criterion	-1.682494	
Sum squared resid	1.014243	Schwarz criterion	-1.603854	
Log likelihood	86.28346	Hannan-Quinn criter.	-1.650676	
Durbin-Watson stat	2.262536			

Breusch-Godfrey Serial Correlation LM Test:

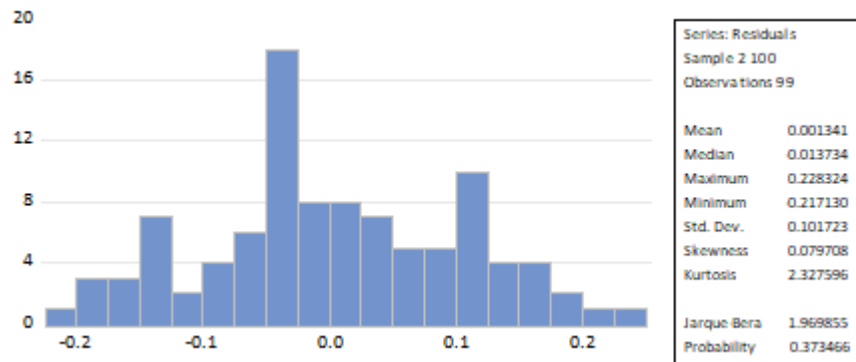
Null hypothesis: No serial correlation at up to 1 lag

F-statistic	2.254379	Prob. F(1,95)	0.1366
Obs*R-squared	2.294843	Prob. Chi-Square(1)	0.1298

Heteroskedasticity Test: White

Null hypothesis: Homoskedasticity

F-statistic	1.018565	Prob. F(6,92)	0.4183
Obs*R-squared	6.166740	Prob. Chi-Square(6)	0.4048
Scaled explained SS	3.862069	Prob. Chi-Square(6)	0.6953



$$y_t = 0,455y_{t-1} + 0,197x_t + 0,312x_{t-1} \quad \text{ср. с уравнением DGP:}$$

$$y_t = 0,5y_{t-1} + 0,2x_t + 0,3x_{t-1}$$



Моделирование в Eviews

Долговременный мультипликатор для модели, которой следуют данные ($y_t = \mu + a_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$):

$$y_t = 0,5y_{t-1} + 0,2x_t + 0,3x_{t-1}$$

вычисляется по формуле $\frac{\beta_0 + \beta_1}{1 - a_1}$. Получаем $\frac{0,2 + 0,3}{1 - 0,5} = 1$ и долговременное равновесие $y_t = x_t$.

Для модели 9 ($y_t = 0,455y_{t-1} + 0,197x_t + 0,312x_{t-1}$) долговременное равновесие: $y_t = 0,934x_t$.

Для модели 0 ($y_t = 0,455y_{t-1} + 0,197x_t + 0,313x_{t-1}$) долговременное равновесие: $y_t = 0,936x_t$.

Для модели 7 ($y_t = 0,448y_{t-1} + 0,432x_{t-1}$) долговременное равновесие : $y_t = 0,783x_t$.



Банк России

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